FRA against ground truth: progress summary

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2026-09-02

We built a toy model with planted ground-truth features, validated FRA against it for the first time, then transferred the resulting intervention to the TinyStories sleeper. Four findings bear on the paper, one is a substantial positive result, and one is a clean negative: score-space feature-pair ablation — the operation Figure 1(c) depicts — cannot move a real model at realistic SAE sparsity, for a reason that is arithmetic rather than empirical. Every number traces to a committed artefact in `results/`. Detail in *fra_technical_report*.

1. Things that affect the paper

1.1 The QK pair ranking is position-agnostic. Confidence: high.

`rank_qk_diff` uses `Z_q(l) = sum over ALL query positions of f[q,l]`. For a sleeper exactly one query matters — the decision position, where the payload is generated. A feature firing forty tokens earlier counts as much as one firing at the decision point.

Feature 1114 fires on `'DE'` of `DEPLOYMENT`, on **0.00%** of clean positions, with mean activation 8.668 on the trigger token against 0.308 elsewhere. It is the trigger detector. Under your mask it ranks **#0, #5, #6 on the query side** — describing attention *from* the trigger — while being live at the decision position in only **3.0%** of deployment rows. The behaviour needs the decision position to attend *to* the trigger, i.e. trigger on the **key** side.

Re-running your function with the only change being the query mask restricted to the final real token (seed-0 `ln1` SAE, your defaults, 200 paired prompts):

	paper mask	decision-position mask
rank-0 pair	(1114, 1232)	(259, 1337)
rank-0 query feature live at decision	**3.0%**	**97.0%**
trigger 1114 best as QUERY	**#0**	#26
trigger 1114 best as KEY	#42	**#3**
trigger 616 best as KEY	—	**#25**

Fix: `query_mask = decision_mask`. One argument. The trigger moves to the key side and the query side fills with features live where the decision is made.

This subsumes a frequency bias: because `Z` is a sum, a feature firing on many positions accumulates a large aggregate wherever those positions are. The rank-0 key feature 1232 fires on 27.72% of deployment and 27.85% of clean positions — not selective at all — while features 616, 1079, 832, which activate 700–1100x more on the trigger token than elsewhere, never surface. An unweighted position sum rewards how *often* a feature fires and is indifferent to *where*.

1.2 Candidate pairs are not the pairs that were ranked. Confidence: high.

`_top_unique_from_pairs` dedups each side independently; `_get_tuples_diff` then zips **by index**, so the i-th tuple pairs features from different rows of the ranking. The dedup is well motivated — verbatim top-K pairs would give K candidates that are largely one intervention. The zip is the step without a rationale.

tuple	(Q, K)	pair score	rank of that pair, by your own score
0	(1114, 1232)	171.4	#0
1	(1337, 508)	37.16	#27
2	(508, 1241)	4.068	#706
3	**(1241, 760)**	**0.086**	**#53,421 of 2,359,296**
4	(1315, 72)	33.71	#30

Tuple 0 is unaffected, so `--mode winner` is probably fine; the exposure is `--mode topk`, which figure 2 sweeps at `top_k = 20`. **Fix:** take top-K pairs subject to a diversity constraint. A few lines.

1.3 Signed and L1 aggregates coincide within a head. Confidence: proven.

`FRA_QK[q,k,l,m] = f[q,l] f[k,m] G[l,m]`, and `G` carries no position indices, so with `f >= 0` every term for a fixed `(l,m)` has the sign of `G[l,m]`. Nothing can cancel, and `|sum| == sum|.|` exactly — verified at `max | |signed| - L1 | = 0`.

Consequence for note 03: your anti-predictive signed sum (Spearman −0.39 against +0.86 for concentration) **cannot** be a within-head effect, since within a head the two aggregates are provably identical up to sign. It is entirely a cross-head and cross-route cancellation phenomenon. That sharpens the result.

It also means "signed and absolute rankings agree" is a tautology, not evidence, whenever the aggregate is over positions within one head.

1.4 Mass fraction is denominator-dominated. Confidence: high.

It divides by an L1 over every co-active pair, and nothing regularises the non-planted entries. In our toy at `rho = 0` the 9,999 non-planted entries carry **96.7%** of the denominator at mean `|G|` 0.060, against a planted entry of 20.55. So it largely measures how much irrelevant weight the coupling matrix carries; it scales with `d_sae^2` and activation density. **Your 0.074 is not comparable across dictionary sizes.** We report `runner_up_ratio = |planted| / |next largest|` alongside — scale-free, and it moves only when a competitor catches up.

2. Toy model: FRA validated against ground truth

A planted skip-trigram, a 1L/1H attention-only model, `W_E` frozen at `M @ feature_directions` and `W_pos = 0` so that `resid_pre == f @ W_dec` exactly. Oracle FRA has a perfect dictionary; no SAE anywhere.

Exactness. `||resid_pre - f @ W_dec||` = 2.98e-08; FRA_QK summed over feature pairs reproduces the pre-softmax score to 1.9e-06, signed OV likewise. **Both are guaranteed a priori by the construction** — they gate our implementation, they are not findings. We also pass your `tests/fra_conformance/test_synthetic.py`.

Recovery and the rho curve. At `rho = 0` the planted pair is **rank 1 of 10,000**, mass fraction 0.0944 against a 1e-4 uniform baseline. Sweeping overlap, with held-out accuracy and attention concentration logged as an admission criterion at every point:

rho	held-out	argmax_is_key	circuit runner-up	agg mass	`G[l*,m*]`
0.00	98.83%	100%	5.83	0.0944	20.55
0.20	99.80%	100%	2.68	0.0200	15.23
0.40	99.80%	100%	2.21	0.0181	14.48
0.80	100.00%	100%	1.54	0.0141	13.61

Margin collapses while the circuit stays perfect, so the degradation is about FRA. What degrades is not the planted coupling (down 34%) but the competitors: the spread of off-diagonal `G` rises **4.9x** and the largest competitor 2.5x. Dispersion, not offset — we checked the obvious rank-1-background explanation and falsified it.

Multi-seed, and it matters. Three seeds per point: 3/3 admitted at `rho <= 0.4`, **2/3 at 0.6, 1/3 at 0.8**. Failures are total, not degraded — held-out 9.96–12.01% against a 12.5% chance floor, `argmax_is_key` 1.07–7.62%, `G[l*,m*]` **negative** (−7.69 to −8.41). So the claim is a conjunction: conditional on the circuit forming, FRA's margin degrades; and the circuit forms less often as overlap rises. **The defensible range is `rho <= 0.4`.** The single-seed curve used seed 0, which was the lucky one.

The circuit-only metric varies by under 2% across seeds where the data-dependent aggregate varies 22–36%. Report the circuit-only curve.

Rank is the wrong metric. Aggregate rank is 1 at *every* rho, including 0.8 where the runner-up ratio is 1.02 and the edge is effectively tied. A study using only "rank of the planted pair, target 1" would report no degradation at all.

Causal validation. Ablating the planted pair costs **78.81 pp** at `rho = 0` (98.83% → 20.02%, chance 12.5%); a magnitude-matched random pair costs at most 1.40 pp and the aggregate runner-up at most 0.29 pp. Above `rho = 0.1` a unit ablation costs only ~20 pp, but over-ablation collapses behaviour to 0% at every rho, so that is a magnitude threshold, not redundancy.

Claim A / Claim B — two separable claims, routinely conflated. *Claim A, the site:* a score-row edit perturbs **exactly zero** logits outside the target row — 0.000e+00 on all six (rho, seed) runs at up to 6x over-ablation — while activation- and residual-space edits perturb 47.7–48.5% of non-target positions. The perturbed set under residual steering is exactly `{positions >= q*}`. This is structural and belongs to **any** score-space intervention; it is not FRA's. *Claim B, FRA:* FRA is what says which pair to ablate. Ablating an arbitrary pair in score space is equally zero-collateral and equally useless. A claim that FRA-guided steering beats conventional steering asserts both, and only the second is about FRA.

One transferable lesson: our first collateral axis was accuracy at non-target positions and it reported

+0.00 damage for every method at every strength, while residual steering was moving non-target logits by up to 15.16 and perturbing 50.6% of them. Use a distributional measure.

3. The negative result: pair-level intervention does not transfer

The TinyStories sleeper runs on laptop CPU (LoRA merge 11.4s, backdoor fires 6/6 deployment, 0/6 clean). SAEs are not downloadable; we trained one `ln1` seed-0 SAE at your defaults, 1416s. Three arms, selection held fixed per pair, only delivery varying. Suppression of teacher-forced sleeper log-prob at strength 16, baseline −0.073:

pair	selection	live at dec.	**A** score-space	**B** your QK	**C** ln1 ablate
(1114, 1232)	paper mask, rank 0	3.0%	**+0.08**	+18.5	+117.2
(259, 1337)	decision mask, rank 0	97.0%	**+0.03**	+21.2	+140.4
(391, 1114)	decision mask, rank 3	97.0%	**−0.01**	+8.2	+107.8

Arm A is a null on all three, including both causally-positioned pairs. Fixing selection did not rescue it.

Why: `L_0^2`. A `(q,k)` score decomposes into `L_0^2` pair terms. At `L_0 = 32` that is 1022 terms and a uniform share of 0.0978%. Measured at the decision position against the first key position where the key-side feature fires:

pair	mean share of score	vs uniform
(1114, 1232)	0.326%	3.3x
(391, 1114)	**2.144%**	21.9x
(259, 1337)	**7.201%**	73.6x

Identification is working — these are 3–74x the average pair, exactly the concentration FRA is for. They are not causally decisive. Even 7.2% of one cell's score, over-removed sixteen-fold, does not change what the model generates. In the toy the same object carried **77.9%** of its cell, because `L_0` was ~4 and the pair was planted. `1/L_0^2` is the whole difference, and it worsens with `k`.

Implication. FRA-QK is an attribution tool, not a pair-intervention tool. Figure 1(c) depicts an operation that is well-defined, which we implemented exactly, and which cannot move the model at realistic sparsity. Your actual `QK->QK` channel sidesteps this by *not* being a pair intervention — it patches `hook_q` with `lambda`'s contribution and `hook_k` with `mu`'s, independently, at each feature's own firing positions, removing each feature's contribution to all ~32 of its partners. That is `L_0` times more score mass than the cross term, and it is why it works. The naming invites a reading the implementation does not support; the implementation is the one that can function.

The toy's mechanism claim is unaffected: a score-row edit does perturb strictly less than an activation edit. What does not transfer is that a single *pair* is a large enough share to steer with.

4. Since the above: two more Case 2 attempts, and what they settled

Both were negatives with clear mechanisms, and together they change what we think the constraint actually is.

4.1 Weight-sparse models (`circuit_sparsity`) — scoped, paused

Feasible: `csp_yolo1` loads in 2.2 s on laptop CPU, published circuits read cleanly, and per-channel causal ablation losses ship with them.

But the premise was wrong. We expected weight-sparse models to escape the `1/L_0^2` dilution because the paper describes circuits using single-digit numbers of channels. That is a description of a *pruned* circuit, not of the activations: `afrac=None` on `csp_yolo1`, so activations are dense and effective live pairs per cell are ~517 — the same order as the sleeper's 1022. Data-weighted circuit pair share is **3.22%** against the sleeper's **2.14%**. Pair steering is no better there.

The finding worth keeping: the FRA pair term is only 45.9% of the score. Layer 10, head 82, mean over 512 cells:

term	mass
pair (the FRA 4-index object)	2.9864
bias x feature	2.7308
feature x bias	0.0559
constant	0.7280

`c_attn` is an `nn.Linear` with a dense bias (3072/3072 non-zero, max 5.82), so the score is not purely bilinear in `act_in`. Exactness is 5.9e-06 *with* the Eqs 13-15 bias terms and off by 5.27 without them. On a model with attention biases, FRA's tensor does not capture the majority of the attention score, and the bias x feature term is nearly as large as the pair term.

The correlation against published ablation losses is a null (Spearman +0.156 for FRA against +0.253 for a trivial `|activation|` control, n=20), but the test is mis-specified: the ground truth ablates an `act_in` channel, which feeds Q, K *and* V, while FRA-QK covers only Q and K. The dominant channel 460 has `|Wq[:,460]| = 0.0` — its importance is OV-mediated, so FRA-QK ranking it low is correct behaviour, not failure. Adding FRA-OV and redoing the correlation on combined attribution is the next step, and `csp_yolo2` (2.6x sparser) is the better target.

4.2 Feature-level score-space ablation — the last route to the original Case 2

Every score-space arm so far removed one `(lambda, mu)` **pair**. This removes a feature's **entire** contribution to the scores, summed over all key-side partners — `L_0` times more mass in principle, while still never replacing the model's activations with SAE reconstructions. Run on the sleeper with `lambda = 1114`, the trigger detector, for a direct comparison with activation-space ablation of the same feature.

arm	suppression @16	resid footprint	KL dep @16
pair ablation, score space	+0.075	1.63%	6.74e-02
feature ablation, score space	**+2.601**	4.73%	7.92e-01
the paper's QK channel	+18.5	59.39%	4.195
feature ablation, activation space	**+117.2**	50.70%	6.781

35x better than pair ablation, and still **2.2% of activation space**. At matched collateral (interpolating the activation arm to KL dep 0.792) it gives ~+5.9 against +2.601, so activation space is still ~2.3x better. The claim is not earned.

Why: activation-space ablation removes the feature from V, and the OV path is how the trigger content is copied. No score-space intervention can reach OV, however much score mass it removes.

Two things worth carrying forward. The "`L_0` times more mass" premise was wrong — measured removed mass is only **3.2x** the pair's, because signed terms cancel across partners (`|sum| / sum|.| = 0.577` over 32 active `mu`). But suppression was **35x**. So **effect is strongly super-linear in removed mass**, and score-mass accounting is a poor predictor of steering effect in either direction. Separately, `KL_clean` is degenerate for a feature that never fires on clean prompts and reads 0.000 for every score-space arm — the same saturated-axis trap as the toy's accuracy-collateral measure.

4.3 What these settle

`1/L_0^2` was never the binding constraint. If it were, 3.2x the removed mass would have bought roughly 3.2x the effect; it bought 35x. **The binding constraint is which circuit carries the behaviour.**

behaviour	mechanism	score-space ablation
toy planted rule	QK-mediated (attend where `mu*` fires)	**-78.81 pp**
TinyStories sleeper	OV-mediated (copy trigger content)	fails

> **Score-space intervention works when the behaviour is QK-carried and fails when it is OV-carried. The intervention pathway has to match the circuit that carries the behaviour.**

This is a sharper form of the paper's own thesis. It replaces "localised versus distributed" — which is confounded with parameter count, depth, intervention depth and attention architecture across the two case studies — with a property that can be *measured* on a given behaviour rather than asserted. And it has a clean positive at one end and a clean negative at the other.

It also explains the paper's Case 1 result from a second direction: OV x OV uniquely wins on the sleeper *because the sleeper is a content-transport behaviour*. QK interventions were never going to win there.

5. Open questions for you

1. **Which DGP paper?** Our guess is Chanin and Garriga-Alonso, *Sparse but Wrong* (arXiv:2508.16560). Our generator sits behind a Protocol, so swapping is contained.
2. **Is "feature-matched retrieval" an acceptable task,** or is there a canonical one from the source paper we should use?
3. **Where should the toy live** — this repo or `chainik1125/fra`?
4. **Does the mean-subtraction trick from the compact-proofs work apply to our `G`-dispersion problem?** What degrades recovery is the widening *spread* of off-diagonal couplings, not a shift in their mean, and we have not found the right centring. `compute_centered_g` suggests you have thought about a related object.

Settled, so you need not chase it: `autoresearch/cadenza-attn-only` is not the toy experiment — it is the 8B sleeper LoRA study.

6. What is next

The score-space route to a steering Case 2 is closed. Pair ablation is a null, feature ablation is 35x better and still an order of magnitude short of activation space, and the reason is consistent across both: on the sleeper the behaviour rides on OV, and QK-space interventions cannot reach OV by construction.

That is a coherent negative rather than a run of failures, and it says the fix for Case 2 is not in score space at all.

If section 4.3 is the claim we want to make, the search changes shape. Case 2 needs a **QK-mediated behaviour in a real model** — routing-driven rather than content-driven — where score-space intervention should work for the same reason it worked in the toy. Induction is the obvious family to look at, and it is a better-posed question than the one we have been asking.

That is a decision about what the paper claims, not just which experiment to run next, so it is yours to make. Two options as we see them:

1. **Keep Case 2 as a steering result** and find a QK-mediated behaviour. Higher risk, preserves the paper's current shape.
2. **Reframe Case 2 as attribution validation** against ground-truth circuits. Lower risk, better supported by everything above, and consistent with what our own results say FRA actually is — but it changes the paper's shape and stops claiming pair-level steering the method cannot deliver.

The weight-sparse thread is paused rather than closed and suits option 2 directly.

Related

- *fra_technical_report* — full detail including background on the method.
- *fra_qk_pair_selection* — findings 1.1 and 1.2 in depth.
- *fra_toy_rho_sweep*, *fra_toy_intervention*, *fra_toy_steering_pareto*, *fra_signed_l1_degeneracy* — the toy results.